

Luc's Statistical Coffee Price Forecast

Monthly Model Report

2026-10-01

Executive Summary

Coffee Price Forecast Summary

Our forecast calls for a modest decline in Arabica prices over the next three months, with the central estimate falling from 342.3 *¢/lb* in month one to 323.3 *¢/lb* by month three. Robusta prices are expected to remain relatively stable, hovering around 182 *¢/lb* across the forecast period. The 80% forecast range spans about 62 *¢/lb* for Arabica and 31 *¢/lb* for Robusta, reflecting greater price volatility in the higher-priced arabica market.

The model's historical accuracy is reasonable for near-term forecasts but degrades over longer horizons. One-month forecasts average 6.1% error for Arabica and 5.6% for Robusta. By three months out, errors rise to 11.4% and 11.9% respectively, which is typical for commodity models. Coverage data is currently unavailable for assessment.

The spread signal between the two varieties is neutral, with a z-score of 0.18. This suggests the Arabica-Robusta price relationship is neither stretched nor compressed relative to historical norms, implying no compelling directional bias between the two markets at present. The mean reversion half-life of 22.7 months indicates any current spread deviations would take roughly six months to correct halfway back to equilibrium.

Current Forecasts

Median (p50) price forecasts and 80% confidence intervals (p10–p90) for the next three months.

Symbol	Horizon	Median (¢/lb)	80% CI (¢/lb)	Band width
Arabica (KC=F)	h=1	342.3	[317.2, 380.8]	63.6
Arabica (KC=F)	h=2	330.8	[306.5, 368.0]	61.4
Arabica (KC=F)	h=3	323.3	[299.6, 359.6]	60
Robusta (RM=F)	h=1	182.1	[167.5, 198.5]	30.9
Robusta (RM=F)	h=2	183.8	[169.0, 200.3]	31.2
Robusta (RM=F)	h=3	181.5	[166.9, 197.7]	30.8

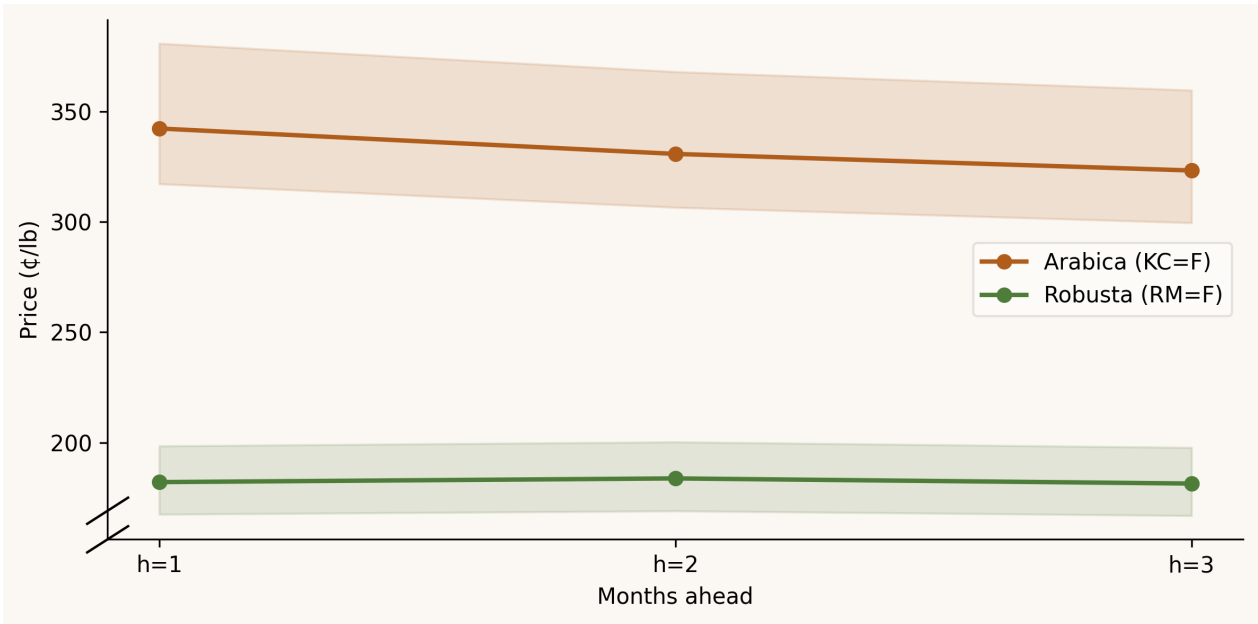


Figure 1: Price trajectory with 80% confidence ribbon

Model Performance

Mean absolute error (MAE), mean absolute percentage error (MAPE), and 80% interval coverage across all walk-forward backtest windows.

Symbol	Horizon	MAE (¢/lb)	MAPE	Coverage 80%
Arabica	h=1	12.8	6.1%	N/A
Arabica	h=2	19.8	9.2%	N/A
Arabica	h=3	25.3	11.4%	N/A
Robusta	h=1	7	5.6%	N/A
Robusta	h=2	11.3	9.1%	N/A
Robusta	h=3	15.1	11.9%	N/A

Coverage 80% target: 80% of forecast ranges should contain the realized price. Values below 80% indicate the model is over-confident.

Recent Forecast Performance

One-month-ahead forecasts versus realized prices over the last three months.

Month	Symbol	Forecast (¢/lb)	Actual (¢/lb)	Error
2026-03	Arabica (KC=F)	312.2	334.1	-6.6% (below actual)
2026-03	Robusta (RM=F)	170.9	176.8	-3.3% (below actual)
2026-02	Arabica (KC=F)	360.8	321.4	+12.3% (above actual)
2026-02	Robusta (RM=F)	192.2	179.6	+7.0% (above actual)
2026-01	Arabica (KC=F)	371.1	363.9	+2.0% (above actual)
2026-01	Robusta (RM=F)	182.2	192.3	-5.2% (below actual)

January 2026

The model forecast Arabica at 371.1 ¢/lb and Robusta at 182.2 ¢/lb, with actual prices settling at 363.9 ¢/lb and 192.3 ¢/lb respectively. Arabica missed by only 2.0 percent on the high side, but Robusta underperformed the forecast by 5.2 percent, indicating divergent behavior between the two varieties. The Robusta underestimation likely reflects stronger-than-expected demand from Asian blenders or production concerns in Vietnam that the model's historical relationships failed to capture.

February 2026

The model significantly overestimated both varieties, forecasting Arabica at 360.8 ¢/lb and Robusta at 192.2 ¢/lb against actual closes of 321.4 ¢/lb and 179.6 ¢/lb, with misses of 12.3 percent and 7.0 percent respectively. The sharp decline across both markets suggests a demand shock or supply relief that statistical models trained on recent volatility could not anticipate. Currency strength in coffee-producing nations or a shift in roaster purchasing patterns ahead of seasonal demand would explain why historical price correlations became less predictive.

March 2026

The model reversed course and underestimated both varieties, forecasting Arabica at 312.2 ¢/lb and Robusta at 170.9 ¢/lb against actuals of 334.1 ¢/lb and 176.8 ¢/lb, missing by 6.6 percent and 3.3 percent respectively. The rebound after February's drop indicates the model treated the prior month's decline as a trend continuation rather than an anomaly. Frost or drought concerns in major growing regions or a rapid reversal in currency movements would explain the sharp recovery that purely backward-looking statistical methods failed to predict.

Arabica–Robusta Spread Signal

The spread model tracks whether Arabica is unusually cheap or expensive relative to Robusta. It measures the **z-score** of the log price ratio — how many standard deviations the current ratio sits from its long-run average. A z-score near zero means the relationship is normal; a z-score beyond ± 2 means a rare dislocation (this happens roughly 5% of the time historically) that has tended to revert. A trade signal fires at $|z| > 2$ and closes once the ratio normalises back to $|z| < 0.5$.

As of	2026-07-01
Current signal	Neutral — ratio within normal range, no edge detected
Z-score	0.18 std devs from average
Half-life	22.7 months — typical time to revert to average
Entry threshold	
Exit threshold	

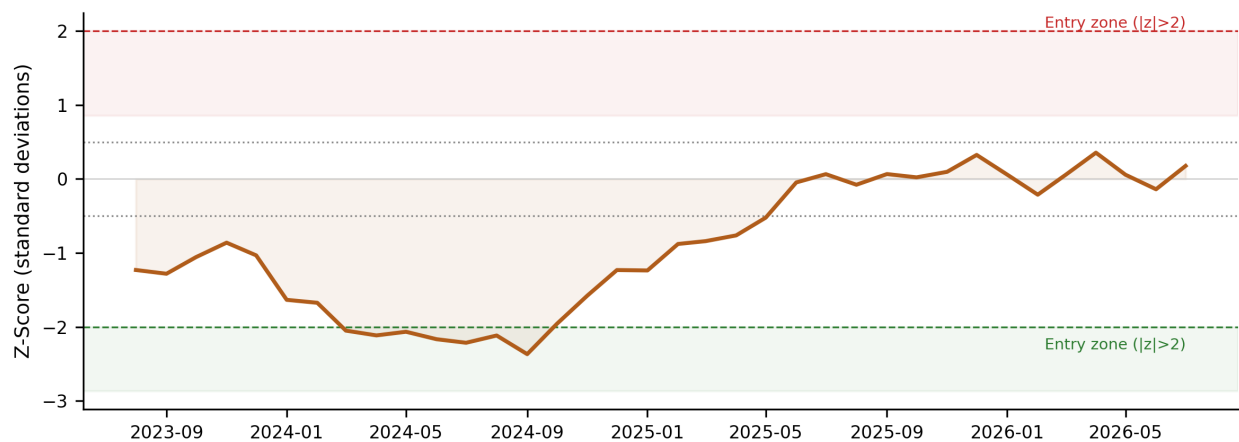


Figure 2: Z-score of Arabica/Robusta log spread — last 36 months. Red zone = Arabica expensive; green zone = Arabica cheap. Dashed lines mark entry thresholds (± 2 std devs); dotted lines mark exit thresholds (± 0.5).

Data & Methodology

Data sources: Arabica (KC=F) and Robusta (RM=F) monthly prices from FRED (PCOFFOTMUSDM, PCOFFROBUSDM). FX drivers (BRL/USD, VND/USD, IDR/USD) from Alpha Vantage. DXY broad index from FRED (DTWEXBGS). Data starts 2000-01-01; monthly series is the mean of daily adjusted closes within each calendar month.

VECM: A two-variable Vector Error Correction Model is fit on log Arabica and log Robusta prices, with BRL, VND, IDR, and DXY as exogenous drivers. Cointegration rank is fixed at 1 (confirmed by Engle-Granger test, $p=0.009$). Lag order is selected by AIC each run. Point forecasts at 1, 2, and 3 months use a naïve FX assumption (hold FX at last observed value).

GAMLSS SHASH: A Sinh-ArcSinh distribution is fit (via R's `gamlss` package) to the VECM forecast residuals, separately per symbol and per volatility regime (Low / Medium / High, defined by rolling 12-month standard deviation). This gives a full predictive distribution. Final price quantiles are $\text{price_qX} = \text{vecm_point} \times \exp(\text{gamlss_residual_qX})$.

Spread signal: An AR(1) model is fit to the log spread series; the expanding z-score triggers entry at $|z| > 2$ and exit at $|z| < 0.5$.

Report generated by the `coffee-forecast` pipeline.